

Spillover Effects between Medicare Programs

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PRELIMINARY

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Abstract

Medicare’s major coverage programs share providers, creating scope for spillovers across segments. We study whether incentives in the Medicare Shared Savings Program (MSSP) spill over from Traditional Medicare to Medicare Advantage (MA). The MSSP can affect MA through two channels. First, *benchmark channel*: because MA benchmarks are tied to local fee-for-service spending, MSSP-induced spending reductions mechanically lower MA plan payments. Second, *provider channel*: because providers treat both Traditional Medicare and MA beneficiaries, MSSP-induced changes in care delivery may spill over directly to MA enrollees.

We focus on the provider channel. To study it, we condition on plan-level MA benchmarks and exploit quasi-experimental variation generated by the 2019 *Pathways to Success* reform, which changed the timing of non-rebasing years by MSSP entry cohort and thereby altered the strength of savings incentives. Combining this policy variation with pre-reform county ACO shares and baseline plan enrollment weights, we construct a Bartik-style instrument for plan-level exposure to MSSP activity.

We find that greater MSSP exposure lowers MA plans’ expected cost of coverage and increases plan generosity. A 1% increase in exposure reduces expected cost per beneficiary by 0.17%, increases rebates by 2.1%, lowers out-of-pocket maxima by 0.37%, and raises the probability of offering a zero-premium plan by 0.57 percentage points. The results imply that MSSP affects MA not only mechanically through benchmarks, but also through provider behavior that spills over across Medicare programs.

Keywords: Health Economics, Medicare Advantage, Medicare Shared Savings Program, Spillover, Bartik Instrument.

1 Introduction

Over the past decade, Medicare has shifted away from pure fee-for-service (FFS) reimbursement toward a patchwork of alternative payment models. Two of the most prominent are Medicare Advantage (MA) and the Medicare Shared Savings Program (MSSP). As of 2025, MA covers roughly half of Medicare beneficiaries through private plans paid a capitated amount per enrollee, while MSSP covers about one-sixth of beneficiaries in traditional Medicare (TM) and aims to lower FFS spending relative to a benchmark. Although these programs are usually analyzed in isolation, healthcare providers typically serve both TM and MA beneficiaries, and practice changes in response to variation in incentives in one segment might spill over to the other.

Spillovers from MSSP to MA Medicare has two main segments: Traditional Medicare (TM) and Medicare Advantage (MA). Under TM, the government sets administered prices for each service and reimburses providers directly on a FFS basis. This payment system is widely viewed as a key driver of excessive utilization and spending. MSSP is designed to curb FFS spending for TM beneficiaries by giving groups of providers—Accountable Care Organizations (ACOs)—a financial incentive to cut some unnecessary spending. ACOs receive a bonus (shared savings) if they keep their annual average per-capita FFS spending for their aligned beneficiaries below a risk-adjusted expenditure benchmark.

Under MA, private plans receive capitated payments to cover the Medicare benefit package. These payments are tied to county-level MA benchmarks that depend mechanically on risk-adjusted FFS spending for TM beneficiaries in the same county. Therefore, MSSP and MA are financially linked: when ACOs succeed in lowering FFS expenditures, they generate shared savings on the TM side, but they also mechanically reduce the MA benchmarks in those counties, tightening the budget constraint faced by MA plans. Lower benchmarks can, in turn, affect plan participation, claims and benefit design. We refer to this spillover that works indirectly through the plan payment rule as the *benchmark channel*.

Secondly, the efficiency-enhancing practice changes (care coordination, chronic disease monitoring, and post-acute care management) developed for ACOs' aligned beneficiaries in response to MSSP incentives can be applied to MA enrollees served by the same organizations. If these changes reduce utilization or shift care toward lower-cost settings, they lower MA claims spending and may induce MA plans to adjust bids and benefit design. We refer to the spillover that works directly through the changes in the ACO providers' practice pattern as the *provider channel*.

Empirical Challenges These two channels are conceptually distinct but empirically intertwined. A policy that strengthens MSSP savings incentives will generally both intensify provider-side care coordination that affects MA enrollees and lower FFS and, in turn, plan payments. Identifying these two channels requires solving several empirical challenges. First, plans’ exposure to MSSP could be endogenous because health plans can self-select into a market with higher ACO penetration, and the latter could be correlated with unobservable market characteristics that also affects the outcome variable. Second, separately identifying the provider channel from the benchmark channel, would require having two instruments that provide exogenous variation in one channel without moving the other. In the plan-level analysis, we do so by conditioning directly on the plan-level MA benchmark and using quasi-experimental variation in MSSP incentives to instrument for plan exposure to ACOs.

Empirical strategy overview For each MA plan p in year t , we construct an endogenous measure of exposure to ACO activity in the plan’s service area, π_{pt}^{ACO} , defined as the enrollment-weighted average of county ACO shares in the counties where the plan operates. This measure can affect plan outcomes both (i) indirectly through benchmarks (since ACO activity can move FFS spending and thus the benchmark) and (ii) directly through provider-side spillovers that change MA claims costs and contracting. Although both channels are important, the objective of this paper is to estimate the spillover effects from the provider channel. To isolate the provider channel, the main specification conditions on the plan-level benchmark B_{pt} and asks whether outcomes vary with π_{pt}^{ACO} holding the plan payment fixed.

To address these sources of endogeneity, we construct a Bartik-style (shift-share) instrument that combines (i) predetermined plan enrollment weights (pre-2019 plan-county enrollment shares), (ii) predetermined ACO penetration across counties (pre-2019 county ACO shares), and (iii) a quasi-experimental time variation in the MSSP incentives. This yields predicted plan exposure to ACO incentives that vary over time due solely to the 2019 nationwide MSSP policy change and cross-sectionally due to pre-2019 ACOs market shares. Following the “exogenous exposure” logic of Goldsmith-Pinkham, Sorkin, and Swift (2020), we use this shift-share component as an instrument for π_{pt}^{ACO} in a 2SLS design and estimate how MA outcomes respond to MSSP exposure, conditional on B_{pt} and market-level covariates.

Dual-role health systems Large health systems operate on both sides of the MSSP–MA linkage. These “dual-role” systems both (i) operate ACOs that manage FFS spending and (ii) sponsor MA plans whose capitated payments are tied to the same county-level FFS spending via the benchmark formula. For such systems, changes that reduce FFS spending

affect their revenue on two margins: they generate larger shared savings bonus payments on the MSSP side, but also mechanically lower MA payments and thus the flow of capitation revenue to their own plans. We incorporate this organizational structure into the analysis by distinguishing exposure attributable to dual-role versus standalone ACOs. This allows us to study whether the incidence of provider-channel spillovers differs when the underlying ACO incentives are generated by dual-role systems that may internalize the benchmark linkage.

Results and Contribution The paper makes three contributions. First, it provides new evidence that MSSP and MA are not separate universes: MSSP penetration is strongly associated with MA plan outcomes at the local level, even after conditioning on benchmarks. Second, it develops a novel Bartik-style instrument that isolates the *provider channel* from the local average treatment effect (LATE) induced by quasi-experimental variation in the MSSP incentive. Third, it highlights how organizational structure—especially dual-role systems that both run ACOs and sponsor MA plans—can shape the incidence and strength of spillovers.

Preliminary estimates are consistent with economically meaningful provider spillovers: higher exposure to ACO activity is associated with plan choices that move in the direction of lower costs of coverage and more generous benefits. Quantitatively, a 1% increase in π_{pt}^{ACO} reduces plans-expected cost per-enrollee by about 0.17% (and by roughly \$1.4 at the sample mean bid of about \$809), increases rebates by about 2.1% (about \$2.1 at the mean rebate of about \$99), and lowers out-of-pocket maxima by about 0.37% (about \$20 at the mean OOP max of about \$5,375). In addition, higher ACO exposure reduces the probability that a plan charges a positive premium (about 0.57 percentage points for a 1% increase in exposure) and is associated with lower plan exit (about 0.40 percentage points for a 1% increase).

2 Related Literature

This paper connects three strands of the literature: evaluations of the MSSP and its fiscal consequences, spillovers between Medicare programs, and the econometrics of shift-share instrumental variables.

MSSP evaluations. A large body of work evaluates whether MSSP ACOs reduce Medicare spending for their attributed beneficiaries. Early studies found modest gross savings concentrated among physician-led ACOs, driven primarily by reductions in spending on post-acute care, outpatient care in hospital-owned facilities, and acute inpatient admissions (McWilliams et al., 2016, 2018; Colla et al., 2016). However, net savings—after subtracting

shared-savings payments—are smaller and sometimes negative (McWilliams et al., 2020). A central concern is that MSSP evaluations may overstate savings because of nonrandom participation and attrition: Markovitz et al. (2019) show that exit of high-cost clinicians can account for a substantial portion of apparent spending reductions, while Ouayogodé et al. (2021) find that favorable provider and beneficiary selection attenuates estimated savings. Our paper does not re-evaluate MSSP savings for attributed beneficiaries; instead, we take ACO activity as given and ask how it spills over to a different segment of Medicare: the MA market.

Spillovers between Medicare programs. A well-established literature documents spillovers from Medicare Advantage to Traditional Medicare. Baicker et al. (2013) show that higher MA penetration reduces hospital utilization for FFS beneficiaries, and Callison (2016) finds that MA penetration lowers treatment intensity across all local Medicare patients. Chernew et al. (2008) document similar effects on total Medicare expenditures, and Feyman et al. (2021) confirm that these spillovers persist in the post-ACA era. Using a randomized payment reform, Einav et al. (2020) provide experimental evidence that payment-model spillovers extend equally to non-targeted patients.

In the reverse direction—from TM programs to MA—Ryan and Markovitz (2023) estimate the mechanical fiscal spillover from MSSP to MA benchmarks: because MA benchmarks are tied to county-level FFS spending, MSSP-induced reductions in FFS spending tighten MA benchmarks, reducing federal payments to MA plans. Within organizations, Navathe et al. (2018) find limited spillovers of ACO participation on post-acute care for non-attributed beneficiaries at participating hospitals, and Post et al. (2019) document modest spillovers from MSSP physician participation to commercial spending.

Most directly related to our paper, Hou et al. (2025) study MSSP spillovers to *non-attributed* FFS beneficiaries using a mover design inspired by Finkelstein et al. (2016). They find that non-attributed beneficiaries who move to areas with greater ACO penetration experience reduced outpatient facility spending and increased physician-services spending, consistent with a shift away from higher-cost facility settings. Their evidence supports the premise that ACO care-coordination practices spill over beyond attributed populations—the same “provider channel” that we study. Our contribution differs in two respects. First, we study spillovers to *MA plans* rather than to non-attributed FFS beneficiaries, examining how provider spillovers translate into changes in plan bids, benefit design, and market participation. Second, we use a shift-share IV design based on the 2019 Pathways to Success reform rather than a mover framework, exploiting quasi-experimental variation in MSSP incentive strength to identify provider-channel effects while conditioning on MA benchmarks.

Shift-share instrumental variables. Our identification strategy relies on a Bartik-style (shift-share) instrument. Two complementary frameworks formalize the conditions under which such instruments are valid. Goldsmith-Pinkham et al. (2020) show that the standard Bartik instrument is numerically equivalent to using the exposure shares as individual instruments in a GMM framework. Under their “exogenous exposure” interpretation, validity requires each share to satisfy a conditional independence (parallel-trends) condition, and they propose Rotemberg weight decompositions and pre-trend diagnostics to assess which shares drive identification. Borusyak et al. (2022) develops an alternative “exogenous shifts” framework in which identification follows from the quasi-random assignment of the shifts, while the shares are allowed to be endogenous; the recent practical guide by Borusyak et al. (2025) synthesizes both approaches and provides implementation checklists. Our instrument combines predetermined plan-county enrollment shares with quasi-experimental variation from the Pathways non-rebasing year schedule, following the logic of Goldsmith-Pinkham et al. (2020): we treat the baseline shares as the source of identifying cross-sectional variation and argue for their exogeneity conditional on plan fixed effects, while the Pathways schedule provides time-varying “shocks” whose timing is pinned down by federal policy rules. Section 5 discusses the identifying assumptions and validation exercises in detail.

3 Institutional Background

3.1 The Medicare Shared Savings Program

In Traditional Medicare (TM), the government pays providers an administered fee f_s for each service s . Let Q_{ist} denote the utilization of service s . The total TM spending for beneficiary i in year t can be written as $Y_{it} = \sum_s f_s \cdot Q_{ist}$. The Medicare Shared Savings Program (MSSP), launched in 2012, aims at reducing unnecessary volume of services by changing organizational practices and through improved care coordination (e.g., chronic care management, high-risk registries, transitional care programs, post-acute management, and reductions in duplicate procedures). Groups of providers that meet specified criteria are allowed to form Accountable Care Organizations (ACOs) and share in savings if they reduce the per-capita fee-for-service (FFS) expenditures for their aligned beneficiaries while maintaining or improving quality.

Financial incentives and the ratchet effect. Each ACO faces an expenditure benchmark that is used to establish whether the ACO generated savings in a given year. If realized spending falls sufficiently below the benchmark, the ACO is rewarded by Medicare a fraction

of the difference between the benchmark and its per-capita spending. This creates a direct financial return to lowering spending for aligned beneficiaries.

The spending benchmark is a weighted average between the ACO’s recent average past spending and the FFS regional average spending. Prior to 2019, ACOs’ participation in the MSSP was organized into agreement periods of three years, and the benchmark is updated every three years to reflect changes in each component. A central challenge of the MSSP design is the *ratchet effect*: successful spending reductions mechanically tighten future benchmarks. Intuitively, if an ACO reduces spending today, rebasing shifts down the benchmark used to measure savings tomorrow, reducing the marginal payoff to sustained savings and weakening dynamic incentives to invest in cost-reducing practice patterns.

Pathways to Success and non-rebasing years. The 2019 *Pathways to Success* reforms represented a major redesign of MSSP intended, in part, to mitigate this ratchet effect. Pathways reorganized participation into five-year agreement periods and introduced *non-rebasing years* within each agreement period. In non-rebasing years, realized spending affects shared savings/losses in the current year but does not tighten the benchmark used in subsequent years. Therefore, ACOs have a stronger marginal incentive to generate savings in those years.

Crucially for this paper, the timing of these non-rebasing years is pinned down by an ACO’s entry cohort and the Pathways policy schedule, and not by the ACO’s choice or local market conditions. Section 5.3 formalizes this schedule and describes how we use it to build plan-level quasi-experimental exposure measures to MSSP incentives.

3.2 Medicare Advantage

Medicare Advantage (MA) is the capitated alternative to Traditional Medicare. Private plans are paid by Medicare a capitated amount per-beneficiary per month to provide coverage for TM Part A and Part B benefits. Plans are differentiated in terms of provider network and coverage, and negotiate with providers the payment rates for each service. In the rest of this section, we will describe how plan payments are set and how this might create linkages between MSSP and MA.

Plan bids, payments, and rebates. Each year, MA plans submit bids that represent the expected per-enrollee cost of providing the Medicare benefit package to a beneficiary of average risk. Let b_{pct} denote plan p ’s bid in county c and year t , and let B_{ct} denote the MA county level benchmark. Medicare pays the plan the lower of the benchmark and the bid, and when a plan bids below the benchmark it receives a rebate equal to a fraction of the

difference that must be returned to beneficiaries through supplemental benefits or reduced cost sharing. In stylized form,

$$\text{Payment}_{pct} = \min\{B_{ct}, b_{pct}\} \quad \text{and} \quad \text{Rebate}_{pct} = \lambda_t (B_{ct} - b_{pct}) \cdot \mathbf{1}\{b_{pct} < B_{ct}\},$$

where λ_t is the statutory rebate share (which depends on plan star ratings in practice). Plans then choose provider networks and plan benefits (premiums, cost sharing, supplemental benefits), and these choices respond to both (i) the payment environment summarized by B_{ct} and (ii) underlying medical costs of covering MA enrollees.

Plan-level benchmarks. The MA benchmark in county c and year t depends on the past FFS spending for the TM beneficiaries in county c , and is computed as

$$B_{ct} = q_{ct} \times FFS_{ct-1} \tag{1}$$

where FFS_{ct-1} is the average per-capita TM FFS spending among TM beneficiaries in the county (risk-adjusted and standardized in the benchmark inputs), q_{ct} is the statutory benchmark factor ¹

Because MA plans typically operate in multi-county service areas, the relevant payment environment for plan p in year t is a plan-level benchmark constructed as the enrollment-weighted average of the county benchmarks in its service area. Let A_{pt} denote the set of counties in which plan p enrolls beneficiaries in year t , and let $w_{pc,t}$ be the share of plan p 's enrollment in year t residing in county c , so that $\sum_{c \in A_{pt}} w_{pc,t} = 1$. The plan-level benchmark is then

$$B_{pt} \equiv \sum_{c \in A_{pt}} w_{pc,t} B_{ct}, \tag{2}$$

where B_{ct} is the county benchmark defined in (1).

3.3 Provider Channel and Benchmark Channel

MA plan outcomes (claims, coverage, generosity, exit) might be affected by changes in ACOs' provider practice behavior through two main channels:

1. **Benchmark channel.** The key institutional linkage to MSSP arises because the MA county benchmark is mechanically tied to county TM FFS spending. Because B_{ct} is proportional to FFS_{ct} , MSSP-induced changes in TM spending translate mechanically

¹There is also a plan quality adjustment, StarBonus_{ct} , that captures the bonus associated with quality ratings.

into changes in MA benchmarks. A rich literature shows that MA plans then adjust entry, coverage, and benefit generosity in response to the induced benchmark shocks.

2. **Provider channel.** When the same provider treats both ACO-assigned beneficiaries and also MA enrollees, MSSP-induced practice changes designed for the former may also affect the MA beneficiaries. If the MA plans take this into account, MSSP incentives might also impact MA claims and, more generally, MA plans outcome Y_{ct} , even when the benchmark B_{ct} is held fixed.

As illustrated in Figure 1, any MSSP policy will in general have both a direct effect on plan-level outcomes through the provider channel and an indirect effect through the benchmark channel. In this paper, we focus on the spillover effects from the provider channel. The empirical strategy below exploits an arguably exogenous change in the MSSP induced savings intensity driven by the non-rebasing years, and controls for the level of the plan-level benchmark, to isolate the provider channel.

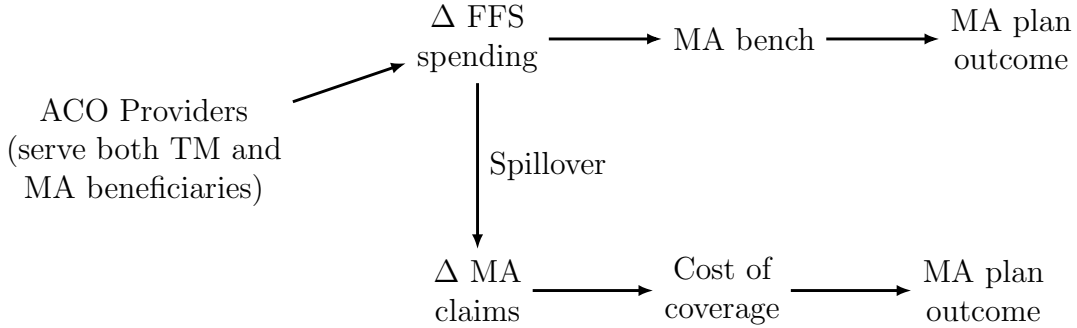


Figure 1: Two channels from MSSP to MA outcomes: benchmark channel and provider channel. The main specification conditions on B_{pt} to isolate the provider channel.

3.4 Dual-Role Health Systems

An important feature of many local markets is that some health systems operate on both sides of the linkage between MSSP and MA. In some cases, the same parent organization (or health system) both (i) sponsors one or more MA contracts and (ii) operates an MSSP ACO that includes its employed physicians and affiliated hospitals. These “dual-role” systems simultaneously influence county-level FFS spending through MSSP and receive capitated MA payments that are tied to that same FFS spending via (1).

This dual role creates a distinctive set of incentives. For an MSSP ACO operated by a system with no MA presence, reductions in FFS_{ct} are unambiguously attractive: they generate shared savings on the FFS side and do not directly reduce any internal capitation

revenue. By contrast, if a health system sponsors MA contracts in the same county, lower FFS spending mechanically reduces the MA benchmark B_{ct} , shrinking the total flow of capitated payments available to the system’s plans. From the perspective of such a vertically integrated system, the marginal benefit of reducing FFS spending reflects a trade-off between: (i) shared savings earned on the MSSP side; and (ii) the implicit reduction in MA revenues arising because MA benchmarks are tied to local FFS.

Recognizing and exploiting this heterogeneity—between markets where MSSP is implemented by standalone provider organizations and markets where MSSP ACOs are operated by MA plan sponsors—is an important part of the empirical analysis (Section 6.2).

4 Data and Descriptive Evidence

The primary analysis sample is a plan-year panel spanning 2013–2023 (coverage years vary by outcome depending on data availability). We construct (i) plan-level outcomes; (ii) county- and plan-level MA benchmarks; (iii) plan-year exposure to MSSP ACOs, π_{pt}^{ACO} ; and (iv) descriptive (non-causal) associations between π_{pt}^{ACO} and MA outcomes. The Pathways-based instrument is constructed and discussed in Section 5.3.

4.1 Main data sources and key variables

MSSP alignment and county-level ACO penetration. From CMS MSSP public-use files and attribution/tabulation products, we observe for each county c and year t the number of FFS beneficiaries aligned to each ACO j , denoted N_{jct} , and the total number of FFS beneficiaries in the county, N_{ct}^{FFS} . We denote the ACO j share in county c in year t with $s_{jct} = N_{jct}/N_{ct}^{FFS}$ and compute the county-level MSSP penetration (“ACOs’ market share”) as the share of FFS beneficiaries aligned to any MSSP ACO:

$$S_{ct} \equiv \sum_{j \in J_c} s_{jct}. \quad (3)$$

where J_c denotes the set of ACOs that have beneficiaries in county c . We also observe ACO entry cohorts, which are used (together with pre-2019 alignment shares) in the shift-share instrument described in Section 5.3.

MA benchmarks at the county and plan level. County benchmarks B_{ct} , statutory benchmark factors q_{ct} , and star-bonus parameters are taken from CMS ratebook files. We construct plan-level benchmarks B_{pt} using plan-county enrollment shares as described in

Section 3.2.

MA plans, enrollment shares, and outcomes. Plan identifiers, service-area counties, enrollment, plan type, and average risk scores are drawn from CMS enrollment and plan characteristics files. We construct plan-county enrollment shares $w_{pc,t}$ from plan-county enrollment counts. Outcomes in preliminary analyses include: (i) (log) average bid; (ii) (log) average rebate; (iii) out-of-pocket maximums (OOP max); and (iv) an indicator for positive premiums, with additional benefit-design variables in ongoing work. Bid and rebate outcomes are obtained from CMS bid pricing tools (BPT) and aggregated to the plan-year level where necessary. Benefit-design variables (e.g., OOP max, premiums, cost sharing) come from CMS benefit and plan characteristics files.

4.2 Plan-level exposure to ACOs

The empirical analysis is conducted at the *plan-year* level. Mirroring the definition of plan-level benchmark, we define plan p 's exposure to MSSP penetration in its service area as the enrollment-weighted average of county ACO penetration:

$$\pi_{pt}^{ACO} \equiv \sum_{c \in A_{pt}} w_{pc,t} S_{ct}. \quad (4)$$

Intuitively, π_{pt}^{ACO} is high when a plan's enrollees are concentrated in counties where a larger fraction of FFS beneficiaries are aligned to MSSP ACOs. In the regressions we use $\ln \pi_{pt}^{ACO}$; in practice we either (i) restrict to positive exposure observations or (ii) use a log-plus-constant transformation, $\ln(1 + \pi_{pt}^{ACO})$, in specifications that include years/plans with near-zero exposure.

4.3 Descriptive Statistics

Figure 2 shows the ACO penetration (S_{c2018}) for the year 2018, which is used in the construction of the ACO Exposure variable (π_{pt}^{ACO}). As seen from the figure there is considerable variation in the ACO penetration across counties. Figure 3 shows the percentage of Medicare beneficiaries enrolled in an MA plan by county for the year 2018. Variation in both variables contributes to the variation in our main variable of interest, ACO exposure (π_{pt}^{ACO}).

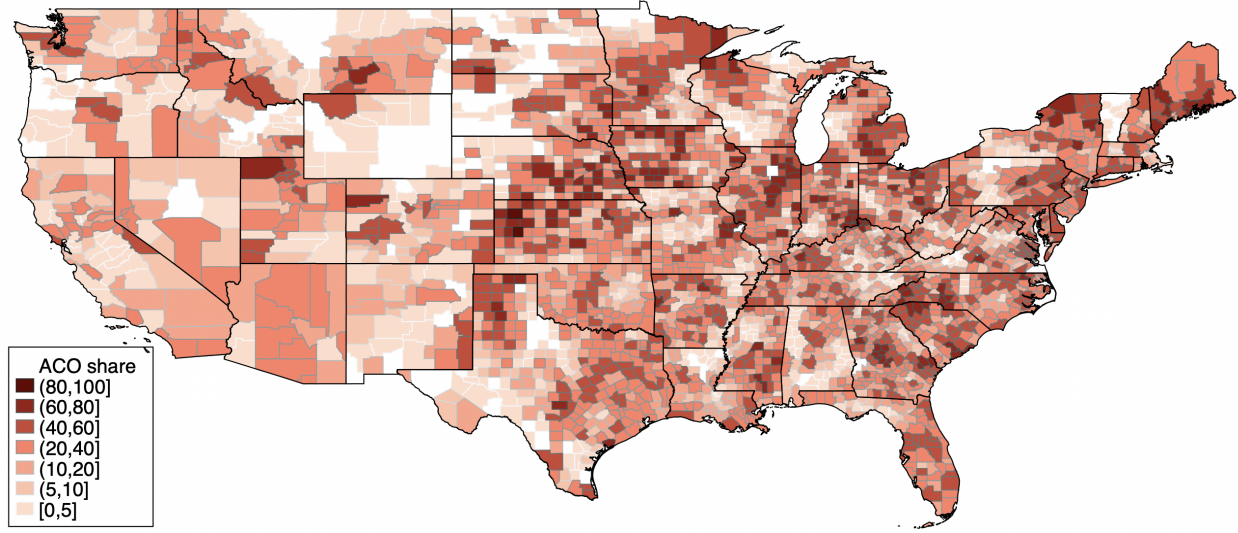


Figure 2: ACO Penetration for 2018 (S_{c2018})

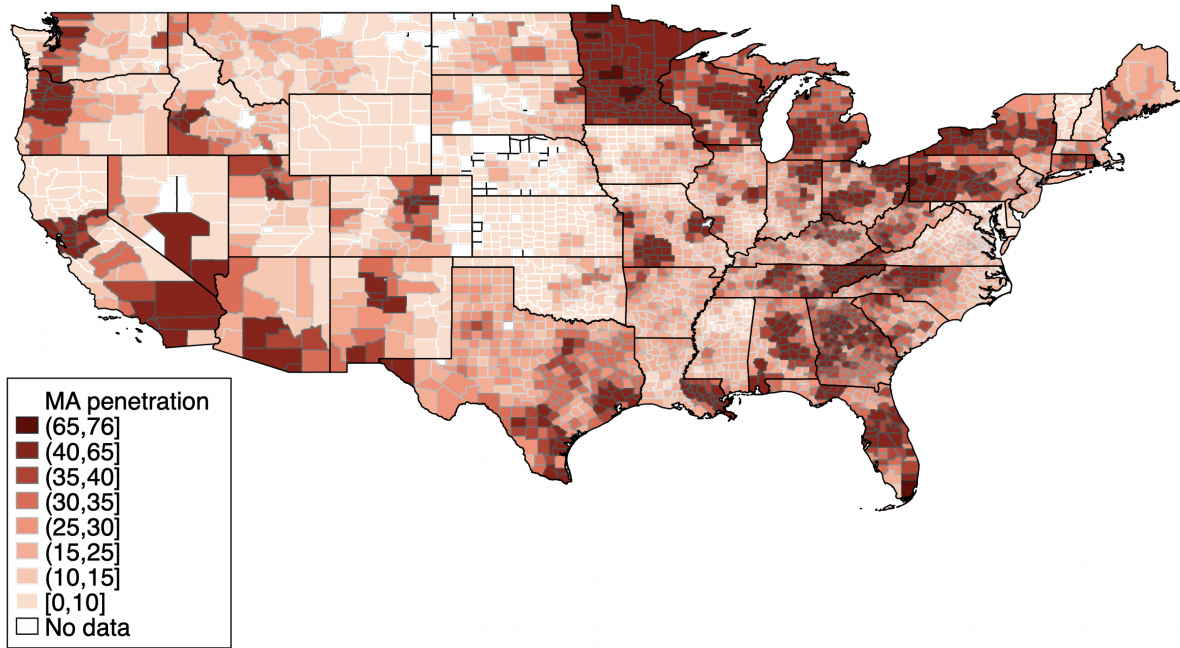


Figure 3: This figure shows the percentage of Medicare beneficiaries enrolled in an MA plan by county for the year 2018.

Appendix B shows the summary statistics tables. Table 5 shows the summary statistics of the ACO exposure (π_{pt}^{ACO}) variable. The ACO Exposure ranges from 0.0 to 0.87 with a mean of 0.298 and a standard deviation of 0.146, indicating significant variation. Table 6 shows the average of the key output variables across plans that face different ACO exposure by quarter. In the raw summary statistics, we observe a positive correlation between ACO exposure and

plan bids. However, this likely reflects selection effects—for instance, ACOs may be more prevalent in high-cost geographic markets—rather than a causal relationship. We address this potential bias using an instrumental variable approach. Regarding rebates, the summary statistics align with our causal findings. Both the raw averages and the instrumental variable estimates indicate a positive relationship showing that plans that are exposed to a higher ACO Exposure are more generous. Table 7 presents the averages for key control variables across plans categorized by ACO exposure quartiles. The data show that benchmarks are higher in counties with greater ACO exposure. Since benchmarks measure average costs in the FFS market, this likely indicates that ACOs disproportionately operate in high-cost geographic markets. Other control variables, such as average risk scores, remain well-balanced across the four quartiles.

5 Empirical Strategy

This section describes the estimating equation, the sources of endogeneity, the construction of the shift-share instrument, and the identifying assumptions. The goal is to estimate the causal effect of plan-level exposure to MSSP ACO activity on MA plan outcomes, operating through the *provider channel*—that is, through changes in provider behavior and care-management practices that affect MA enrollees directly, net of the mechanical benchmark channel.

5.1 Estimating Equation and Parameter of Interest

Let Y_{pt} denote an outcome for MA plan p in year t (e.g., bids, rebates, premiums, OOP max). The baseline specification is:

$$Y_{pt} = \alpha_p + \lambda_t + \theta \ln \pi_{pt}^{ACO} + \beta B_{pt} + X'_{pt} \gamma + \varepsilon_{pt}, \quad (5)$$

where α_p are plan fixed effects, λ_t are year fixed effects, and X_{pt} includes controls such as average risk scores, plan-type indicators, and area-level demographic and healthcare supply characteristics. In some specifications, we additionally absorb contract, state, and HRR fixed effects and HRR-by-year interactions to control for time-varying regional shocks.

The coefficient θ is the object of interest. Conditioning on B_{pt} plays a central role in the interpretation. As shown in equation (2), the plan-level benchmark B_{pt} is a sufficient statistic for the payment environment facing the plan: it summarizes the county-level FFS spending, statutory benchmark factors, and star-rating bonuses that jointly determine capitated payments, rebates, and the budget constraint under which plans design benefits. By holding

B_{pt} fixed, the specification absorbs any effect of MSSP exposure that operates through the benchmark channel ($\text{MSSP} \rightarrow FFS_{ct} \rightarrow B_{ct} \rightarrow \text{plan behavior}$). Any remaining association between $\ln \pi_{pt}^{ACO}$ and Y_{pt} must therefore operate through non-benchmark channels—primarily through provider-side spillovers, as depicted in Figure 1.

Intuitively, θ asks: among plans facing the *same* benchmark, do those whose providers have greater MSSP exposure behave differently? If ACO care-coordination infrastructure (high-risk registries, transitional care programs, post-acute management) spills over to MA enrollees and lowers the cost of coverage, we expect $\theta < 0$ for bids and out-of-pocket costs, and $\theta > 0$ for rebates.

5.2 Endogeneity

The key regressor π_{pt}^{ACO} is the enrollment-weighted average of county-level ACO penetration across the plan’s service area:

$$\pi_{pt}^{ACO} = \sum_{c \in A_{pt}} w_{pc,t} S_{ct}, \quad (6)$$

where $w_{pc,t}$ is the share of plan p ’s enrollment in county c in year t , and S_{ct} is county c ’s realized ACO penetration. Both components are endogenous, and each creates a distinct identification challenge.

Endogenous ACO penetration. County-level ACO penetration S_{ct} can be correlated with local demand conditions, provider market structure, or health system consolidation trends that independently affect MA plan outcomes. For example, counties with more integrated delivery systems may both attract ACOs and offer lower-cost care environments for MA enrollees. If integrated markets have lower MA costs for reasons unrelated to MSSP, OLS would conflate provider spillovers with pre-existing market characteristics, likely biasing $\hat{\theta}$ toward finding cost reductions even if the true provider spillover were zero.

Endogenous plan geography. The enrollment weights $w_{pc,t}$ change over time as plans adjust their service areas, marketing strategies, and provider networks. If plans strategically expand into counties where ACO presence reduces costs (or retreat from counties where tightening benchmarks make operations less profitable), these geographic adjustments can mechanically move π_{pt}^{ACO} in ways that are jointly determined with plan outcomes. This source of endogeneity is especially relevant because plan service-area decisions are made simultaneously with bidding and benefit-design choices.

5.3 Instrument Construction

To address both sources of endogeneity, we construct a shift-share (Bartik) instrument that combines predetermined exposure weights with quasi-experimental time variation from the 2019 Pathways to Success reform.

Source of exogenous variation: the non-rebasing year schedule. As described in Section 3.1, the Pathways reform introduced five-year agreement periods with designated *non-rebasing years*. In these years, spending reductions do not mechanically tighten the ACO’s future benchmark, strengthening the marginal incentive to cut costs. Crucially, the timing of non-rebasing years is determined by an ACO’s *entry cohort* and the national agreement-period calendar—not by local market conditions or plan decisions. We define:

$$K_{jt} \equiv \mathbf{1}\{\text{ACO } j \text{ is in a non-rebasing year at time } t\}, \quad (7)$$

and treat K_{jt} as a policy-determined indicator. Figure 4 in Appendix C, shows when the non-rebasing years occur for each cohort of ACOs. For the set of ACOs that enter the MSSP prior to 2019, we consider the timing of the non-rebasing years to be exogenous as it is determined by the cohort year and the changes introduced with the 2019 Pathway policy.

Step 1: Baseline plan footprints. Let $t_0 = 2018$ denote the last pre-Pathways year. We fix plan-county enrollment weights at their t_0 values: $w_{pc}^{base} \equiv w_{pc,t_0}$, and set $w_{pc}^{base} = 0$ for counties not served by plan p in t_0 . Let A_p^{base} denote the set of counties with $w_{pc}^{base} > 0$. By freezing the plan footprint, we eliminate endogenous geographic adjustments from the instrument.

Step 2: Baseline county-by-ACO alignment shares. Let S_{cj}^{base} denote the share of TM assignable beneficiaries in county c aligned to ACO j at the baseline year t_0 . This captures the pre-reform “dose” of each ACO in each county.

Step 3: County-level non-rebasing-year exposure. We construct predicted county-year exposure to anti-ratchet incentives as:

$$P_{ct} \equiv \sum_j S_{cj}^{base} K_{jt}. \quad (8)$$

All time variation in P_{ct} is driven by the Pathways schedule K_{jt} ; the cross-sectional pattern is pinned down by the baseline ACO composition of each county. In counties where baseline

ACOs happen to belong to entry cohorts whose non-rebasing years arrive early, P_{ct} rises sooner; in counties dominated by later-cohort ACOs, the increase comes later.

Step 4: Plan-level shift-share (Bartik) instrument. We aggregate county exposure to the plan level using baseline enrollment weights:

$$Z_{pt} \equiv \sum_{c \in A_p^{base}} w_{pc}^{base} P_{ct} = \sum_{c \in A_p^{base}} w_{pc}^{base} \left(\sum_j S_{cj}^{base} K_{jt} \right). \quad (9)$$

The instrument Z_{pt} removes both sources of endogeneity identified above: (i) the county ACO composition is frozen at baseline, eliminating confounding from post-reform changes in local ACO penetration; and (ii) the plan’s geographic footprint is frozen at baseline, eliminating confounding from strategic service-area adjustments. The only source of time variation is the nationally determined non-rebasing year schedule, which switches on and off mechanically for each ACO entry cohort.

5.4 Two-Stage Least Squares

We estimate (5) by 2SLS, using Z_{pt} (or its log transform) as an excluded instrument for $\ln \pi_{pt}^{ACO}$. The first stage is:

$$\ln \pi_{pt}^{ACO} = \alpha'_p + \lambda'_t + \mu Z_{pt} + \beta' B_{pt} + X'_{pt} \rho + \eta_{pt}. \quad (10)$$

We expect $\mu > 0$: plans whose baseline service areas load more heavily on ACO cohorts currently in non-rebasing years should have higher realized ACO exposure, because non-rebasing year incentives intensify care coordination and help retain aligned beneficiaries.

Under the identifying assumptions of Section 5.2, $\hat{\theta}$ is a LATE of Pathways-induced variation in MA plan exposure to MSSP spending reduction incentives, operating through the provider channel net of the MA benchmark changes. In other words, it captures the effect of plan-level ACO exposure on MA outcomes for the subpopulation of plans whose exposure is shifted by Pathways-induced variation in MSSP incentives, conditional on the benchmark environment B_{pt} .

5.5 Identifying Assumptions

In order to correctly identify the LATE of stronger MSSP incentives induced by the non-rebasing years, two conditions must hold:

Relevance. The instrument Z_{pt} must predict $\ln \pi_{pt}^{ACO}$ conditional on included controls and fixed effects. We assess this through the Kleibergen-Paap first-stage F -statistic, reported for each specification. Because we cluster standard errors at the plan level, we report Kleibergen-Paap rather than Cragg-Donald F -statistics.

Exclusion. Conditional on plan and year fixed effects, B_{pt} , and X_{pt} , the instrument Z_{pt} must be uncorrelated with ε_{pt} . This requires that non-rebasing year timing—interacted with predetermined ACO alignment shares and plan enrollment weights—affects MA plan outcomes *only* through its effect on realized ACO exposure π_{pt}^{ACO} .

Following the logic of Goldsmith-Pinkham et al. (2020), our identifying assumption is that the baseline exposure shares—the plan-county enrollment weights w_{pc}^{base} and county-ACO alignment shares S_{cj}^{base} —are exogenous conditional on the included fixed effects and controls. Because both are measured in the pre-Pathways period and the panel includes plan fixed effects, the assumption requires that these baseline shares are uncorrelated with *changes* in unobserved determinants of MA plan outcomes after 2018, conditional on observables. This would be violated if, for example, plans that happened to have high baseline exposure to early-cohort ACOs were on differential outcome trends for reasons unrelated to non-rebasing year incentives.

We view the exclusion restriction as plausible for two reasons. First, non-rebasing year timing is mechanically determined by entry cohort and calendar time; ACOs cannot choose or manipulate when they enter a non-rebasing year. Second, we condition on B_{pt} , which absorbs the mechanical benchmark channel through which MSSP-induced FFS spending changes affect MA payments. Any residual correlation between Z_{pt} and MA outcomes must therefore operate through non-benchmark channels. Section 7 probes these assumptions through a battery of validation exercises.

6 Results

This section reports 2SLS estimates of the provider-channel effect of MSSP exposure on MA plan outcomes, followed by instrument validation and robustness exercises.

6.1 Plan-year outcomes: bids, rebates, and plan benefits

Table 4 reports 2SLS estimates of equation (5) for the main plan outcomes. All specifications include plan and year fixed effects, condition on the plan-level benchmark B_{pt} , and control for average risk scores and area-level characteristics. Standard errors are clustered at the plan

level. In richer specifications, we additionally absorb contract, state, HRR, and HRR-by-year fixed effects and weight by the log of the 65-plus service-area population.

Bids. Higher instrumented exposure to ACO activity is associated with lower plan bids, both in levels and in logs (Table 1, columns (1)–(2)). The log specification implies an elasticity of about -0.176 (s.e. 0.033): a 10% increase in π_{pt}^{ACO} lowers bids by roughly 1.8% (about \$14 at the mean bid). The level specification is consistent with this magnitude: the coefficient of -135.6 (s.e. 26.2) implies a \$13.6 reduction for a 10% increase in exposure. This pattern is consistent with ACO care-coordination practices lowering the cost of covering MA enrollees. The underlying mechanism can be understood by analogy to the evidence in [Hou et al. \(2025\)](#), who find that non-attributed FFS beneficiaries in high-ACO-penetration markets shift from hospital outpatient departments to lower-cost physician office settings. If this shift in practice patterns extends to MA enrollees treated by the same providers, plans’ expected claims costs fall, and competitive pressure translates lower costs into lower bids.

Rebates. Higher ACO exposure is associated with higher rebates (Table 1, columns (3)–(4)). In logs, the estimated elasticity is about 2.12 (s.e. 0.53), implying that a 10% increase in exposure raises rebates by about 21% (roughly \$21 at the mean rebate). In levels, the coefficient is 117.5 (s.e. 23.2), implying an increase of about \$11.8 for a 10% exposure increase. Since rebates equal a fraction of the gap between the benchmark and the bid, $\lambda_t(B_{ct} - b_{pct})$, this is the mirror image of the bid result: lower bids, holding the benchmark fixed via the B_{pt} control, mechanically increase the bid–benchmark spread and thus the rebate returned to beneficiaries through supplemental benefits.

Out-of-pocket maximums. Plans with greater ACO exposure offer lower out-of-pocket maximums (Table 1, columns (5)–(6)). In logs, the coefficient is about -0.373 (s.e. 0.155), implying that a 10% increase in exposure reduces OOP maxima by roughly 3.7% (about \$200 at the mean OOP max). The level specification implies a similar magnitude: -2568 (s.e. 636), corresponding to about \$257 for a 10% increase in exposure. These estimates suggest that provider-channel savings are partially passed through to beneficiaries in the form of more generous cost-sharing protections. Plans that face lower expected claims costs can afford to cap beneficiary exposure at a lower level without increasing financial risk.

Premiums, copays, and supplemental benefits. Higher ACO exposure is associated with a lower probability of charging a positive premium (Table 2, column (1)). The point estimate of -0.574 (s.e. 0.251) implies that a 10% increase in exposure reduces the probability

of a positive premium by about 5.7 percentage points (relative to a mean of about 26%). We also find evidence of reductions in cost sharing: primary-care copay indicators fall with exposure (Table 2, column (2); coefficient -0.412 , s.e. 0.214), while specialist copays move in the same direction but are imprecisely estimated (Table 2, column (3)). Taken together, these results complete the picture: provider spillovers reduce plans' cost of coverage, and plans respond along multiple benefit dimensions—lower bids, lower OOP caps, lower premiums, and (in some dimensions) richer cost-sharing terms.

Plan exit. Higher ACO exposure is associated with a lower probability of plan exit (Table 2, column (4)). The coefficient of -0.398 (s.e. 0.158) implies that a 10% increase in exposure reduces exit by about 4.0 percentage points, which is economically large relative to a mean exit rate of about 8.7%. This is consistent with ACO spillovers improving plan viability by reducing claims costs: plans operating in higher-ACO-exposure service areas face a more favorable cost environment and are less likely to find continued participation unprofitable.

Table 1: Plan-year outcomes and MSSP exposure (2SLS)

	(1) Bid	(2) ln Bid	(3) Rebate	(4) ln Rebate	(5) OOP Max	(6) ln OOP Max
ln π^{ACO}	-135.6*** (26.18)	-0.176*** (0.0327)	117.5*** (23.22)	2.123*** (0.531)	-2568.1*** (636.0)	-0.373** (0.155)
ln MA Bench	600.9*** (26.08)	0.751*** (0.0318)	44.24* (26.45)	2.194*** (0.488)	3358.4*** (623.4)	1.109*** (0.212)
ln Risk Score	65.35*** (9.246)	0.0965*** (0.0113)	8.939 (8.121)	-0.649*** (0.171)	-718.1*** (192.1)	-0.261*** (0.0753)
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
HRR FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17552	17552	17552	17552	17552	17552

Notes: Each column reports a 2SLS plan-year regression. The endogenous regressor is $\ln \pi_{pt}^{ACO}$, plan-level exposure to MSSP penetration as defined in (4), instrumented by the Pathways-based shift-share variable Z_{pt} from (9). “Bench Plan” is the plan-level benchmark B_{pt} from (2). We control for a rich set of market-level demographics and healthcare supply characteristics. Standard errors clustered at the plan level are in parentheses. [First-stage F -statistics to be added.] *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The economic magnitudes are consistent with meaningful provider spillovers: the bid reductions imply that MA plans' expected per-enrollee costs decline when their provider base has stronger MSSP incentives, even holding the payment environment fixed. These findings

Table 2: Plan-year outcomes and MSSP exposure (2SLS)

	(1) Premium>0	(2) PCP Copay	(3) Specialist Copay	(4) Exit
π^{ACO}	-0.574** (0.251)	-0.412* (0.214)	-0.190 (0.171)	-0.398** (0.158)
Bench Plan	-0.356* (0.187)	0.0350 (0.173)	0.289* (0.158)	-0.0739 (0.120)
Risk Score	0.127*** (0.0308)	-0.273*** (0.0303)	-0.433*** (0.0316)	0.00892 (0.0128)
Plan FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
HRR FE	Yes	Yes	Yes	Yes
Observations	15353	15366	15366	15366

Notes: Each column reports a 2SLS plan-year regression. The endogenous regressor is $\ln \pi_{pt}^{ACO}$, plan-level exposure to MSSP penetration as defined in (4), instrumented by the Pathways-based shift-share variable Z_{pt} from (9). We control for a rich set of market-level demographics and healthcare supply characteristics. “Bench Plan” is the plan-level benchmark B_{pt} from (2). Standard errors clustered at the plan level are in parentheses. [First-stage F -statistics to be added.] *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

complement the beneficiary-level evidence in Hou et al. (2025), who document similar site-of-care substitution effects on the FFS side; our results show that the same provider-level changes feed through to MA plan pricing and benefit design.

6.2 Heterogeneity by Dual-Role Systems

We explore how the provider-channel spillover varies with organizational structure. As described in Section 3.2, some health systems operate on both sides of the MSSP-MA linkage: they run MSSP ACOs *and* sponsor MA plans whose capitated payments depend on the same county-level FFS spending. For these “dual-role” systems, reducing FFS spending generates shared savings on the TM side but also mechanically lowers MA benchmarks, creating a tension that may dampen their incentive to invest aggressively in care coordination.

We implement two tests. First, we classify plans as vertically integrated (VI) when the plan’s parent organization owns or controls major provider assets in its service area (e.g., hospitals or large physician groups). We then estimate versions of (5) that interact $\ln \pi_{pt}^{ACO}$ with the VI indicator to test whether plans with tighter provider integration exhibit different magnitudes of provider-channel spillovers.

Second, we decompose county-level exposure P_{ct} into components attributable to dual-

role systems versus standalone ACOs (systems without MA plan sponsorship in the same market), and aggregate these to the plan level using baseline weights. This yields separate instruments P_{pt}^{dual} and $P_{pt}^{standalone}$ that allow us to compare how plan outcomes respond to MSSP exposure generated by dual-role versus standalone provider organizations. If dual-role systems internalize the benchmark externality and invest less in care coordination as a result, spillovers should be weaker when driven by dual-role ACOs than by standalone ACOs.

Table 3: Plan-year outcomes and MSSP exposure (2SLS) with Dual Role ACOs

	(1) ln Bid	(2) ln Rebate	(3) ln OOP Max
ln π^{ACO}	-0.389*** (0.0981)	2.260*** (0.816)	-1.206*** (0.448)
ln $\pi^{ACO} \times$ DR share	0.733** (0.342)	-3.089 (3.222)	2.277 (1.411)
ln MA Bench	0.761*** (0.0382)	1.973*** (0.270)	1.014*** (0.223)
ln Risk Score	0.0798*** (0.0168)	-0.357*** (0.127)	-0.210*** (0.0689)
Plan FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
HRR FE	Yes	Yes	Yes
Observations	17538	15081	17456

Notes: Each column reports a 2SLS plan-year regression. The endogenous regressor is $\ln \pi_{pt}^{ACO}$, plan-level exposure to MSSP penetration as defined in (4), and its interaction with the dual role share DR_{pt} . The instrumental variables are Z_{pt} as defined by (9), and Z_{pt} interacted with DR_p^{base} . We control for a rich set of market-level demographics and healthcare supply characteristics. “Bench Plan” is the plan-level benchmark B_{pt} from (2). Standard errors clustered at the plan level are in parentheses. [First-stage F -statistics to be added.] *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Plan-year outcomes and MSSP exposure (2SLS) with Dual Role ACOs

	(1) Premium>0	(2) Copay	(3) Specialist Copay	(4) Exit
$\ln \pi^{ACO}$	-0.807* (0.419)	-0.318 (0.337)	0.384 (0.367)	0.533 (0.460)
$\ln \pi^{ACO} \times \text{DR share}$	0.933 (1.452)	0.888 (1.188)	-1.104 (1.106)	-1.794 (1.476)
$\ln \text{MA Bench}$	-0.163 (0.126)	-0.356** (0.139)	-0.224* (0.123)	-0.0817 (0.137)
$\ln \text{Risk Score}$	0.273*** (0.0588)	-0.0551 (0.0508)	0.0140 (0.0419)	0.182*** (0.0610)
Plan FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
HRR FE	Yes	Yes	Yes	Yes
Observations	15542	17538	17538	17538

Notes: Each column reports a 2SLS plan-year regression. The endogenous regressor is $\ln \pi_{pt}^{ACO}$, plan-level exposure to MSSP penetration as defined in (4), and its interaction with the dual role share DR_{pt} . The instrumental variables are Z_{pt} as defined by (9), and Z_{pt} interacted with DR_p^{base} . We control for a rich set of market-level demographics and healthcare supply characteristics. “Bench Plan” is the plan-level benchmark B_{pt} from (2). Standard errors clustered at the plan level are in parentheses. [First-stage F -statistics to be added.] *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Conclusion

This paper studies whether incentives created by the Medicare Shared Savings Program (MSSP) spill over to Medicare Advantage (MA) through providers’ practice patterns. The motivating idea is that providers commonly treat both Traditional Medicare and MA patients. If MSSP contracts change how providers deliver care for Traditional Medicare beneficiaries, some of those changes may carry over to MA enrollees and affect MA plans’ costs, benefit design, and exit decisions.

A central challenge is that a plan’s observed exposure to ACO activity is not random. Local ACO penetration can evolve with market-level shocks that also affect MA outcomes, and plans can endogenously adjust their geographic enrollment mix and networks in ways that mechanically change measured exposure. To address these concerns, the empirical strategy leverages the 2019 *Pathways to Success* reform and uses the policy-driven timing of non-rebasing years to construct predicted exposure that holds fixed pre-policy plan footprints and pre-policy county-level ACO composition. This design is intended to isolate quasi-experimental variation in the intensity of MSSP incentives within the areas where plans operated before the reform.

The preliminary findings point to economically meaningful provider spillovers. Higher instrumented exposure to ACO activity is associated with plan behavior that is consistent with lower costs of coverage and partial pass-through of savings to beneficiaries. In particular, the patterns in bids, rebates, and multiple benefit dimensions suggest that when providers face stronger incentives to reduce Traditional Medicare spending, the resulting changes in practice patterns may also lower expected costs for MA enrollees treated by the same providers. The estimates also point toward improved plan viability in higher-exposure settings, consistent with a more favorable cost environment.

We also show how our findings are affected by the “dual-role” health systems—organizations that both participate in MSSP through affiliated ACOs and operate MA plans. These large organizations internalize the negative impact of FFS spending reductions in the form of lower payments for their sponsored MA plans.

These results have two broader implications. First, they suggest that MSSP may generate benefits beyond the aligned Traditional Medicare population by altering provider behavior in ways that spill over across Medicare programs. Second, they underscore the importance of separating the provider channel from the benchmark channel when evaluating interactions between Traditional Medicare reforms and MA outcomes.

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Appendix A

Summary Statistics

Table 5: Summary statistics of ACO Exposure

	Mean	Median	25th Perc.	75th Perc.	Min	Max	SD
ACO Exposure	0.298	0.303	0.196	0.400	0.000	0.871	0.146

Table 6: Summary statistics by ACO Exposure

	Q1	Q2	Q3	Q4	Total
Average Bid	790.3 (109.6)	806.5 (108.8)	812.0 (103.1)	829.8 (105.2)	808.7 (107.8)
Premium > 0	0.295 (0.456)	0.246 (0.431)	0.225 (0.418)	0.274 (0.446)	0.261 (0.439)
Average Rebate	91.66 (81.38)	96.85 (81.49)	104.2 (79.12)	104.8 (75.42)	99.01 (79.72)
Out-of-Pocket Max	5106.4 (1825.2)	5553.1 (1645.0)	5471.4 (1647.1)	5393.9 (1587.8)	5375.4 (1692.8)
PCP copay	1.364 (0.481)	1.316 (0.465)	1.296 (0.457)	1.299 (0.458)	1.320 (0.467)
Specialist Copay	1.289 (0.454)	1.253 (0.435)	1.229 (0.420)	1.185 (0.389)	1.242 (0.428)
Exit	0.0916 (0.288)	0.0774 (0.267)	0.0968 (0.296)	0.0838 (0.277)	0.0874 (0.282)
Observations	4783	4545	4216	4008	17552

The above table shows the mean of each of the outcome variables across the 4 quarters of ACO Exposure. The standard deviations are shown in parentheses.

Table 7: Summary statistics of control variables by ACO Exposure

	Q1	Q2	Q3	Q4	Total
Benchmark	884.3 (99.10)	912.3 (105.6)	924.0 (90.66)	944.9 (89.63)	914.9 (99.26)
Risk Score	1.141 (0.387)	1.156 (0.383)	1.195 (0.417)	1.140 (0.346)	1.158 (0.385)
Insurer HHI	8.005 (0.355)	7.977 (0.318)	8.007 (0.264)	8.000 (0.272)	7.997 (0.307)
Hospital HHI	8.294 (0.704)	8.455 (0.488)	8.518 (0.431)	8.617 (0.431)	8.463 (0.546)
Household Income	64822.8 (17328.1)	64147.4 (13978.8)	65376.7 (15196.8)	66971.8 (14599.8)	65271.6 (15413.4)
Inpatient Medicare Payment	9.769 (0.107)	9.785 (0.107)	9.777 (0.0868)	9.778 (0.0813)	9.777 (0.0969)
Outpatient Medicare Payment	2.059 (0.0238)	2.063 (0.0227)	2.068 (0.0210)	2.072 (0.0200)	2.065 (0.0225)
Pop share above 65	0.153 (0.0315)	0.153 (0.0354)	0.161 (0.0327)	0.167 (0.0340)	0.158 (0.0339)
Observations	4783	4545	4216	4008	17552

The above table shows the mean of each of the control variables across the 4 quarters of ACO Exposure.
The standard deviations are shown in parentheses.

Appendix B

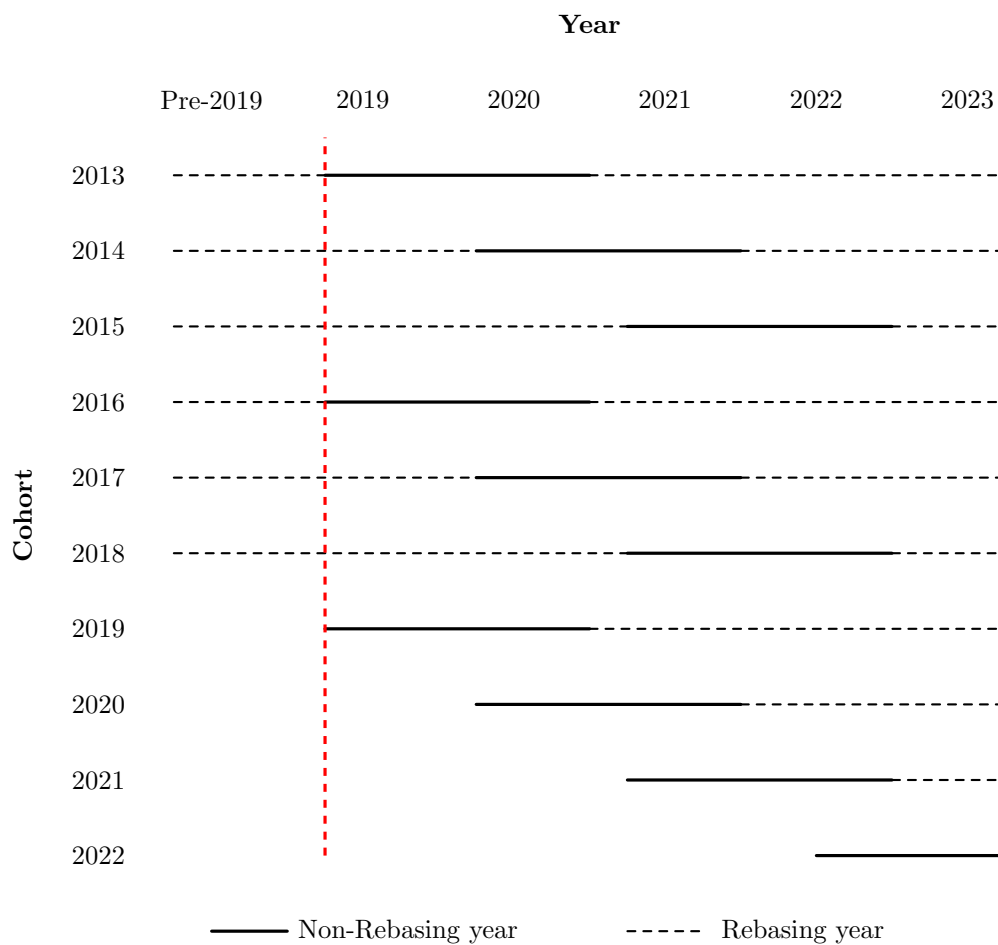


Figure 4: This figure shows when the non-rebasing years occur for each entry cohort.

Appendix C

Instrument Validation and Robustness

We implement several diagnostic exercises to probe the validity of the identification strategy. Throughout, the excluded instrument is the Pathways-based shift-share variable Z_{pt} defined in (9).

(i) Event-Study Analysis

The most direct test of the identifying assumptions is whether the instrument predicts changes in MA outcomes *before* the Pathways reform took effect. If the instrument is valid, there should be no systematic relationship between baseline Pathways exposure and pre-reform plan outcomes, since non-rebasing year incentives did not exist before 2019.

To implement this test, we summarize each plan’s exposure with a time-invariant “dose” constructed as the post-2018 average of the time-varying instrument:

$$D_p \equiv \frac{1}{T_{\text{post}}} \sum_{t \in \text{post}} Z_{pt}. \quad (11)$$

We then estimate a dose-response event study:

$$Y_{pt} = \alpha_p + \lambda_t + \sum_{k \neq -1} \beta_k \left(\mathbf{1}\{t - 2019 = k\} \times D_p \right) + X'_{pt} \gamma + \varepsilon_{pt}, \quad (12)$$

where $k = -1$ (i.e., 2018) is the omitted baseline year. The identifying prediction is that *lead* coefficients are approximately zero: $\beta_k \approx 0$ for $k \leq -2$.

Intuition. To see concretely why this test is informative, consider two MA plans, Plan *A* and Plan *B*, that are identical in 2018: they serve similar counties, face the same benchmarks, and offer comparable benefits. The only difference is that Plan *A*’s service area happens to be concentrated in counties where the dominant ACOs entered MSSP in 2013, while Plan *B*’s counties have ACOs that entered in 2016. Under the Pathways schedule, the 2013-cohort ACOs reach their first non-rebasing years before the 2016-cohort ACOs do, so after 2019 Plan *A* receives a higher “dose” D_p than Plan *B*.

The event study asks: were Plans *A* and *B* already on different *outcome trajectories before 2019*, when neither cohort faced non-rebasing year incentives? If the pre-2019 coefficients $\beta_{-2}, \beta_{-3}, \dots$ are indistinguishable from zero, it suggests that the two plans were evolving in parallel before the Pathways schedule activated. Any post-2019 divergence in bids, rebates, or

benefits can then be more credibly attributed to the differential non-rebasing year exposure, rather than to pre-existing differences between markets where early versus late ACO cohorts operate.

We estimate (12) both for the main MA plan outcomes and for $\ln(\pi_{pt}^{ACO} + 1)$ as an intermediate outcome. In the latter case, the expected pattern is no differential trend before 2019, followed by a divergence as the Pathways schedule begins differentially increasing ACO effort for high-dose plans.

(ii) Benchmark Robustness

Even if the instrument has some effect on benchmarks, we can assess whether the 2SLS estimate of the provider channel is sensitive to how flexibly we control for B_{pt} . We estimate three alternative specifications:

1. *Quadratic benchmark control:* add $(\ln B_{pt})^2$ to the main specification. This allows the relationship between benchmarks and plan outcomes to be nonlinear—for instance, if benchmark changes have larger effects on plan behavior in low-benchmark markets where margins are thinner.
2. *Year-interacted benchmark slopes:* replace the single coefficient on $\ln B_{pt}$ with year-specific slopes $\mathbf{1}\{t\} \times \ln B_{pt}$. This allows the pass-through of benchmarks to plan outcomes to evolve over time, as it may in practice due to changing statutory rebate shares and ACA-era benchmark adjustments.
3. *Benchmark-decile fixed effects:* replace $\ln B_{pt}$ with indicators for year-specific benchmark deciles. This is the most flexible specification, effectively comparing plans within narrow benchmark bins and eliminating any concern about functional-form misspecification.

Appendix D

Event Study Plots

Figure 5 to Figure 12 show the results from the event study analysis described in Appendix A. We plot the estimated values of β_k for $k \in \{-4, 3\}$, where 2018 ($k = -1$) is treated as the base year since it is the year that precedes the 2019 MSSP policy change.

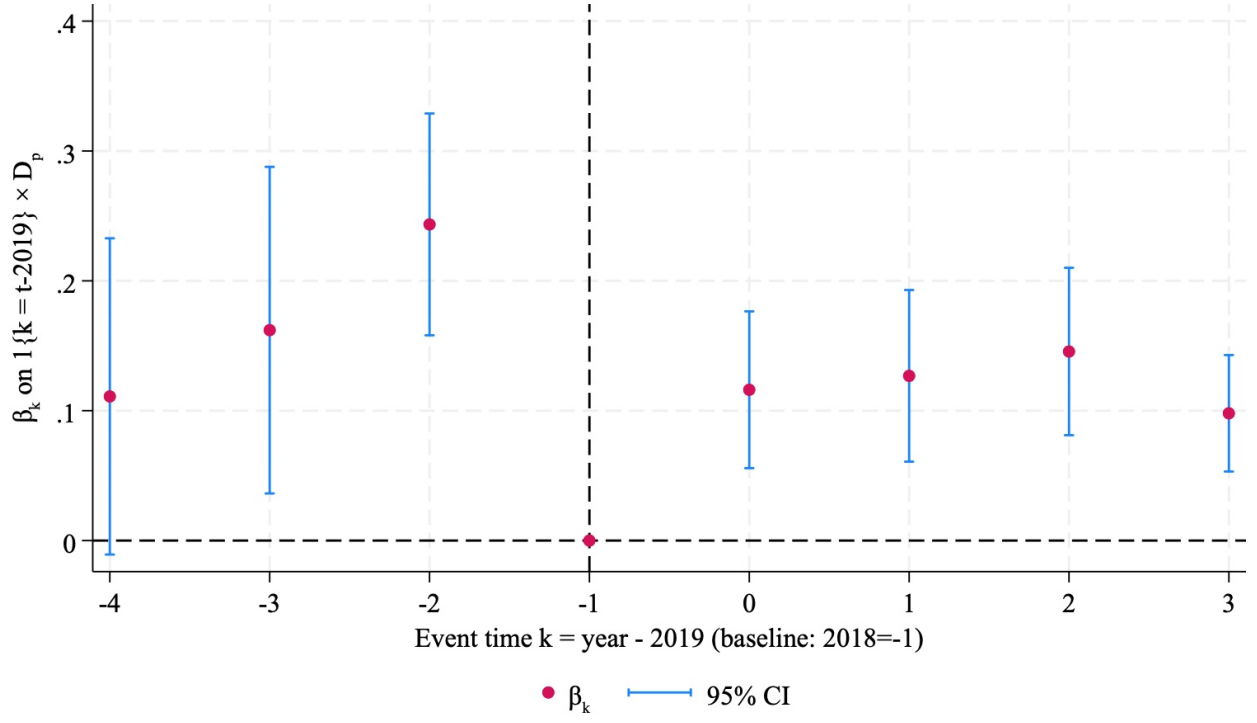


Figure 5: Event-study: log average bid.

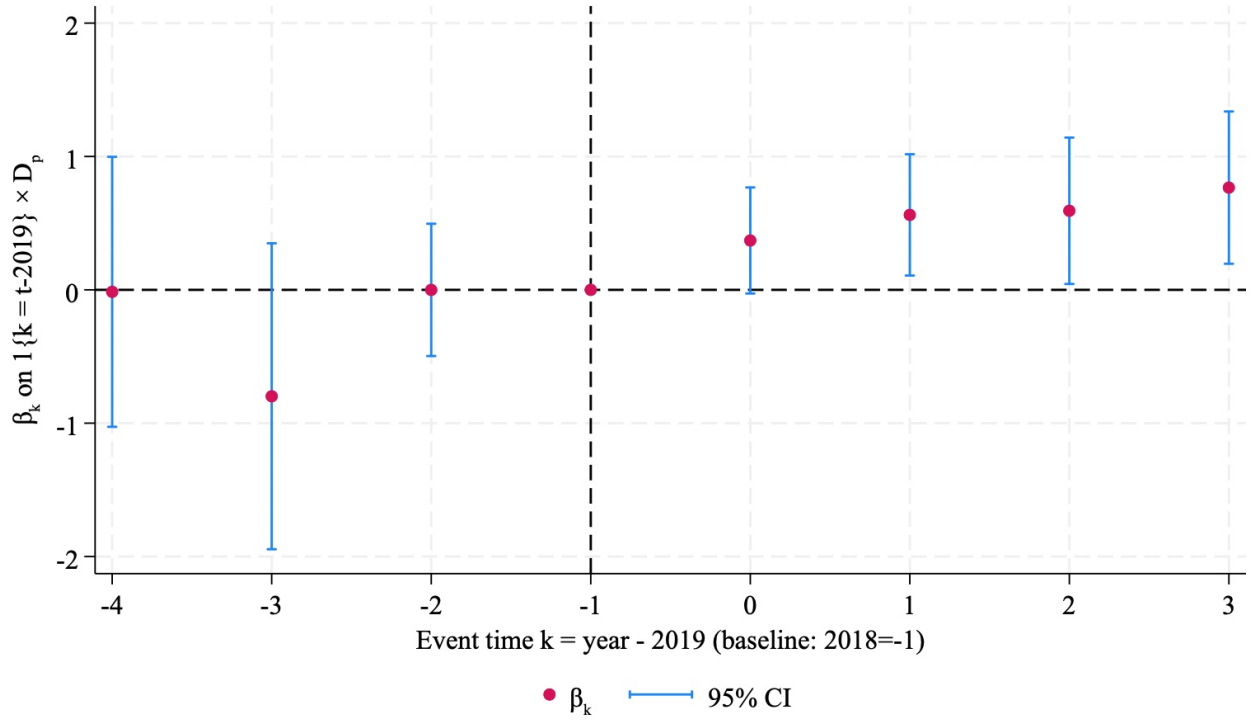


Figure 6: Event-study: log average rebate.

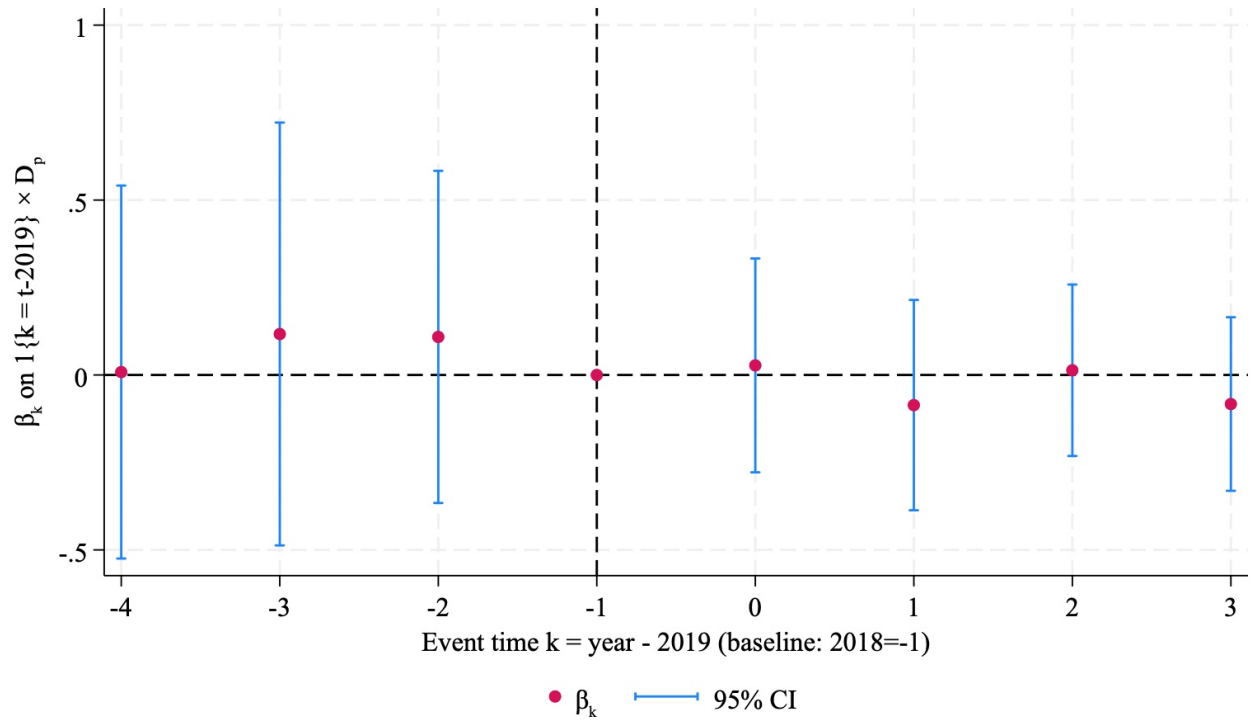


Figure 7: Event-study: plan exit indicator.

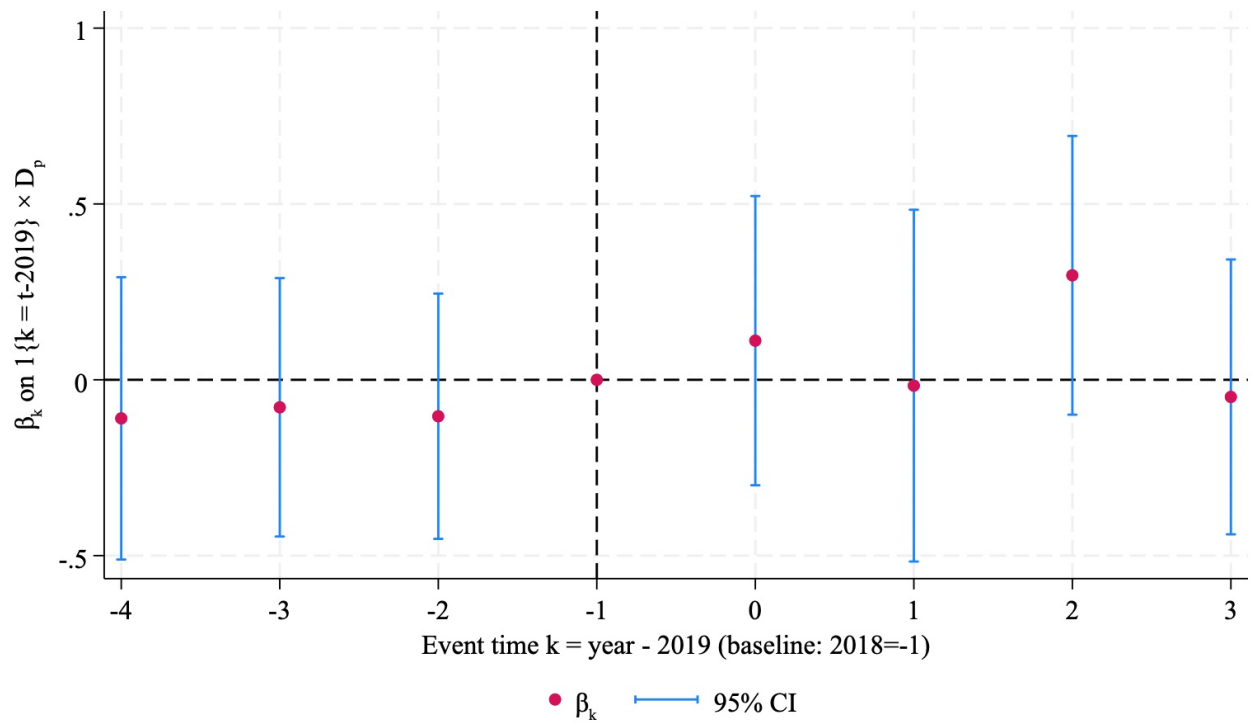


Figure 8: Event-study: log out-of-pocket maximum and exit.

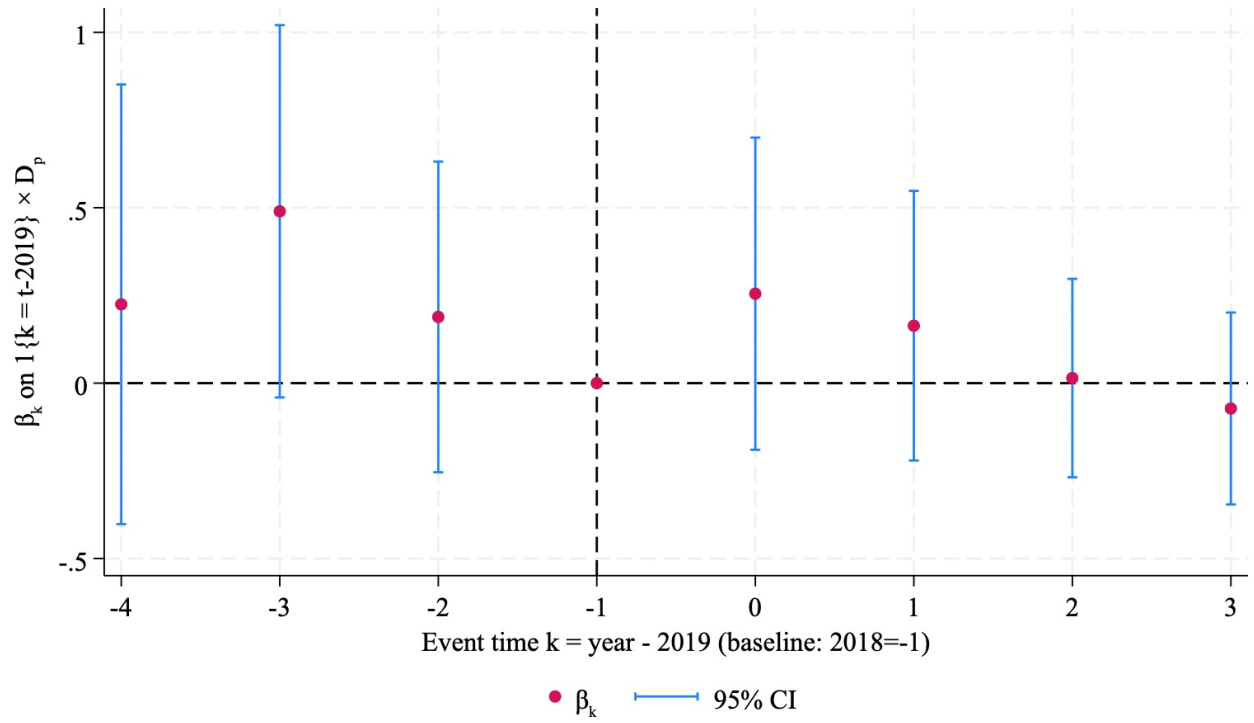


Figure 9: Event-study: Positive Premium indicator

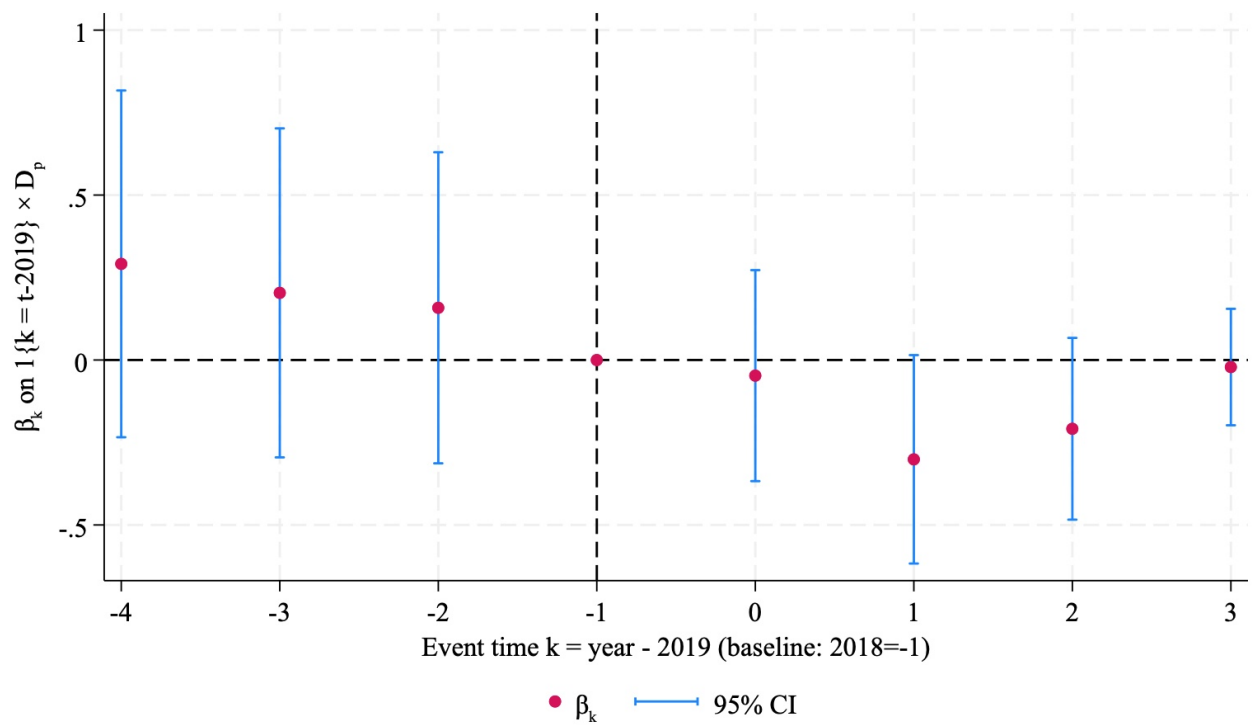


Figure 10: Event-study: PCP copay indicator.

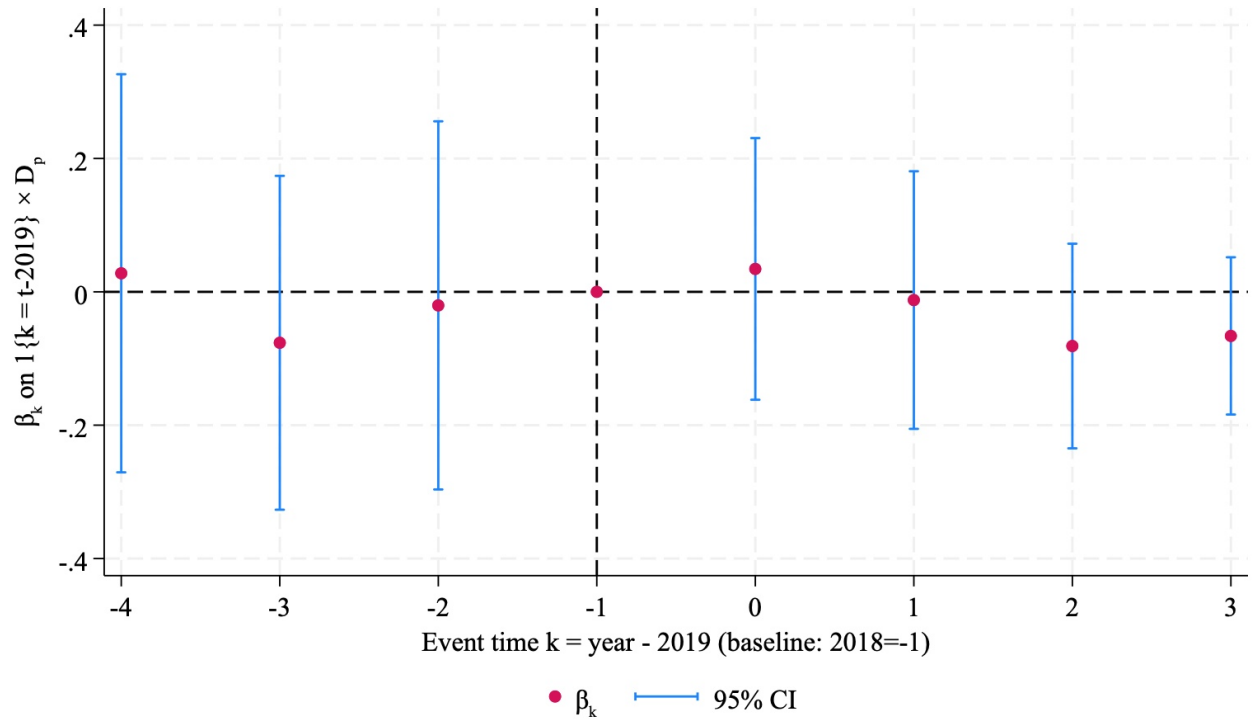


Figure 11: Event-study: Specialist copay indicator indicator.

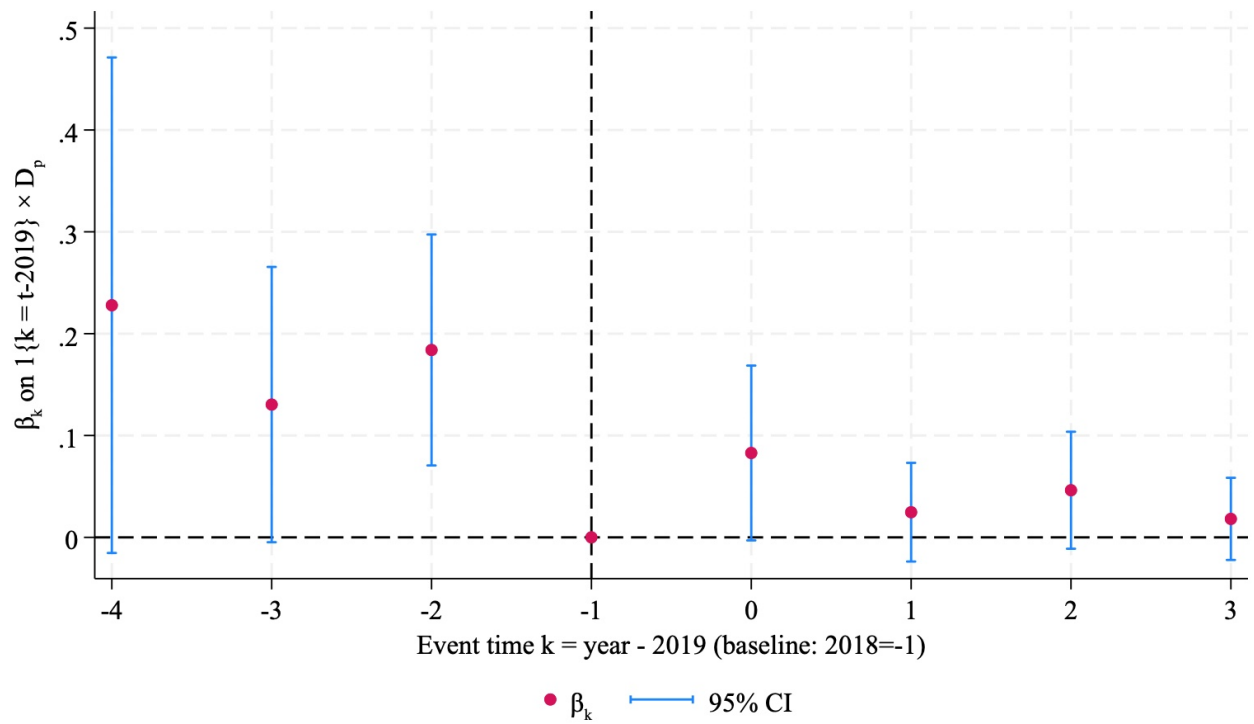


Figure 12: Event-study: Supplemental benefit indicator.